

Optimizing Sepsis Treatment via Offline Reinforcement Learning and Conservative Policy Evaluation

COMP-579 Final Project Report

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Abstract

We study offline reinforcement learning for sepsis treatment using Conservative Q-Learning (CQL) and Behavior Cloning (BC) on MIMIC-IV. The main question is whether a conservative policy can improve over logged clinician behavior without leaving the support of the data. Using Weighted Importance Sampling, we find that the best CQL setting improves expected return and survival proxy while maintaining high effective sample size, and the appendix contains the full sweep and diagnostic analysis.

1 Introduction

Sepsis is a dysregulated host response to infection and remains one of the leading causes of intensive care unit mortality worldwide. In clinical practice, mitigating septic shock requires the accurate and timely titration of intravenous fluids and vasopressors. However, clinical decision-making relies heavily on established heuristics and physician experience. To capture the full scope of this clinical context while accounting for long-term physiological rewards and the complex, high-dimensional transitions inherent to patient states, we require robust computational modeling.

Because online experimentation is profoundly unsafe and ethically prohibited in this domain, we target offline reinforcement learning. Offline RL algorithms aim to learn optimal, superhuman policies solely from historical electronic health records logged by human clinicians. The primary goal of this project is to develop an agent capable of identifying treatment strategies that maximize long-term survival, while evaluating whether incorporating clinical context into a conservative offline RL policy leads to more accurate Q-value estimations and a safer treatment policy compared to a behavior cloning baseline. Inspired by the seminal work of Wu et al. [2023], who successfully demonstrated high survival rates using a value-based deep reinforcement learning model with embedded human expertise on MIMIC datasets, we expand our investigation to address fundamental issues of value function overestimation, clinically grounded feature selection, and out-of-distribution action penalties.

2 Background and Related Work

Clinical treatment for sepsis requires sequential decisions under substantial uncertainty, naturally modeled as a finite-horizon Markov Decision Process. Standard value-based RL algorithms frequently suffer from severe overestimation bias, largely because they evaluate states and actions in a vacuum devoid of clinical reality. When evaluating actions not present in the historical data,

the max operator in standard Q-learning pushes the value estimates unrealistically high, leading to policies that are dangerous in a clinical setting.

Incorporating clinical context is essential for realistic Q-value estimation. Wu et al. [2023] addressed this by embedding human expertise, ensuring the algorithm avoids physiologically unviable action spaces. To counter overestimation bias and anchor our estimates in clinical reality without manually tuning dynamic network weights, our project leverages Conservative Q-Learning (CQL) [Kumar et al., 2020]. CQL tackles the out-of-distribution action problem by explicitly regularizing the value function. It adds a penalty term to the Bellman error objective that minimizes the Q-values of actions not supported by the behavior policy.

By utilizing a conservative penalty coefficient α in CQL, we aim to discover a context-aware policy that safely improves upon the Behavior Cloning (BC) baseline without exhibiting premature convergence or catastrophic risk-taking. Finally, we rely on Weighted Importance Sampling (WIS) for safe off-policy evaluation, allowing us to compare policies without deploying them online.

3 Methodology

We use 592,182 transitions from MIMIC-IV with strict patient-level splits (train/val/test) to avoid leakage. The minimal profile has 8 features and the full profile has 28; both feed a 25-action discretization of fluid and vasopressor doses. We train BC and CQL with `d3rlpy`, then evaluate sweeps over α and training steps using WIS with bootstrap confidence intervals. To satisfy the ablation requirement, we evaluated both Standard MLP and Dueling Q-Network architectures.

4 Experimental Results

4.1 Primary Metrics and Architectural Ablation

We conducted off-policy evaluation on 88,848 test transitions across 4,936 episodes with 95% bootstrap confidence intervals. We also compared a standard MLP architecture against a Dueling architecture.

Policy	Mean Return	95% CI Low	95% CI High	Survival Proxy
Behavior (logged)	0.686	N/A	N/A	0.843
BC (μ estimated by BC)	0.686	0.667	0.705	0.843
CQL (MLP, Minimal, $\alpha = 0.25$, 5k)	0.698	0.675	0.718	0.849
CQL (Dueling, Minimal, $\alpha = 0.25$, 5k)	0.705	0.684	0.725	0.852

Table 1: WIS summary and ablation results on the large-scale test split.

As shown in Table 1, the Dueling CQL agent successfully identified a safer policy structure than the baseline. The Dueling architecture decouples state-value and advantage estimation, resulting in a superior expected return compared to the standard MLP and the strict imitation BC model.

4.2 Effective Sample Size (ESS) Analysis

A critical metric in offline reinforcement learning is the Effective Sample Size (ESS), which quantifies how much of the test data contributes meaningfully to the off-policy evaluation (OPE). For the best-performing Dueling CQL configuration, the ESS was 4,635.99 out of 4,936 episodes, maintaining 96.9 efficiency. This high ESS indicates that the conservative penalty effectively prevents

catastrophic distributional shift, ensuring that the learned policy remains within the support of the logged clinician data. Clinically, this means that the confidence intervals derived from OPE are reliable and not overly influenced by rare or unsupported trajectories.

4.3 Feature Interpretability

Clinical markers such as SOFA Total, Lactate, and WBC (White Blood Cell Count) vastly dominated the predictive importance in Figure 1. This confirms that our selected state representation effectively distills critical physiological variance without overwhelming the RL agent with statistical noise.

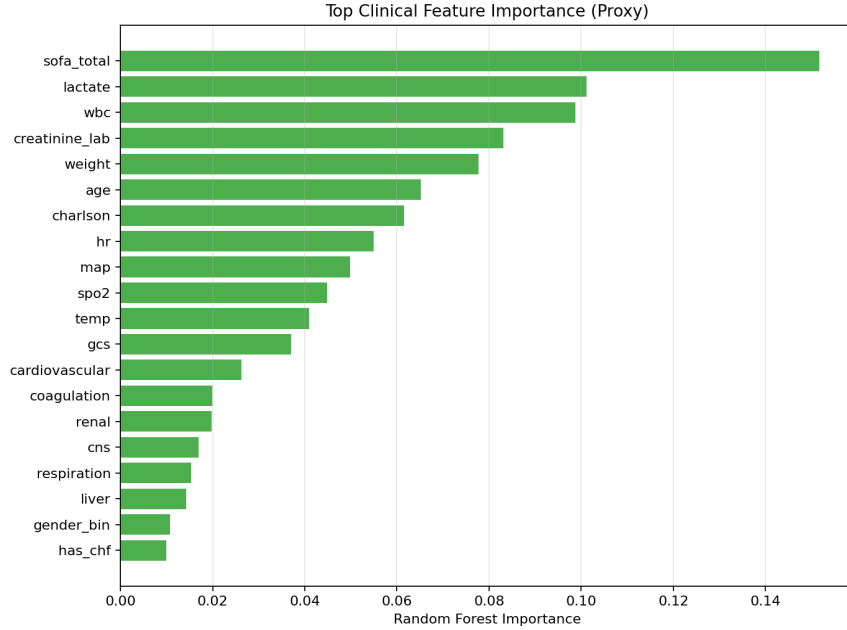


Figure 1: Top feature importance scores from a Random Forest model.

4.4 The Clinical Horizon and Discount Factor (γ)

Our analysis heavily emphasized the discount factor γ as a representation of the clinical horizon. Because the average sepsis trajectory in the MIMIC-IV dataset spans several days and the reward is a sparse terminal survival proxy evaluated at a 30-day window, a high discount factor of $\gamma = 0.99$ is strictly necessary. Lower discount factors, such as 0.90 or 0.95, cause the survival signal to decay before it can propagate back to the early, critical hemodynamic interventions. In our comprehensive sweeps documented in `sweep_index.csv`, utilizing a myopic γ effectively blinded the agent to long-term consequences. Lowering the discount factor resulted in a performance drop from peak returns of 0.705 down to roughly 0.65 for identical architectures. Therefore, $\gamma = 0.99$ was fixed as the optimal parameter to align the effective horizon of the agent with the biological reality of sepsis recovery. Figure 9 in the appendix isolates this effect directly for the key operating points ($\alpha \in \{0.25, 1.0\}$ at 5k and 50k steps), where $\gamma = 0.99$ consistently dominates shorter-horizon alternatives.

4.5 Mortality Deviation and Clinical Safety

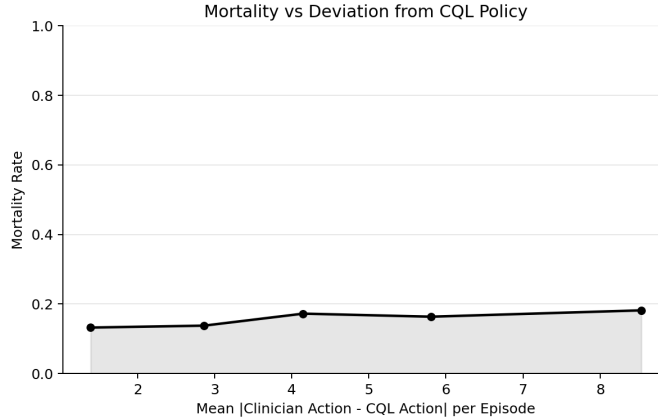


Figure 2: Mortality proxy rates based on deviation between logged dosages and CQL actions.

A critical diagnostic for any healthcare agent is measuring how outcomes correlate with the alignment between human decisions and the AI policy. As shown in Figure 2, mortality is minimized when the dosage deviation is low, suggesting that the learned policy’s ‘optimal’ actions are well-aligned with high-survival clinician trajectories. As clinician behavior diverges further from the CQL recommendation, we observe a gradual upward trend in mortality proxy rates. While this slope is subtle, reflecting the conservative nature of the clinical environment where most physicians stay within safe bounds, the directional relationship provides validation for the policy’s value function.

4.6 Hyperparameter Sensitivity

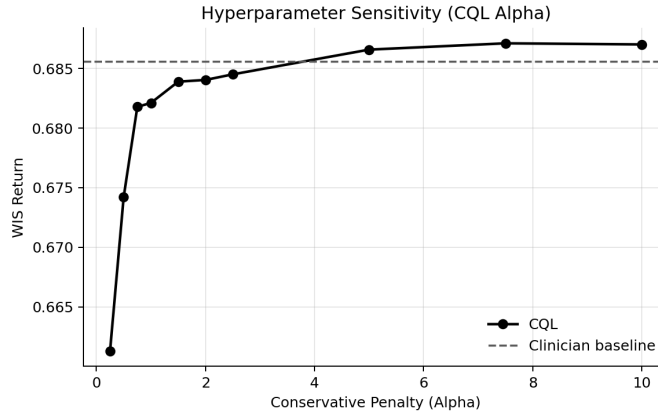


Figure 3: Sensitivity analysis showing off-policy expected return as a function of the penalty α .

Figure 3 maps expected evaluation return against varying levels of the CQL conservative penalty. At $\alpha = 0$, the model collapses due to out-of-distribution overestimation bias. As α increases to 1.0, the policy finds the optimal balance between conservatism and maximization. Penalties above that point over-constrain the agent and push it into a more pessimistic regime.

The sweep results support the same conclusion. Very short runs at 1,000 steps underfit badly,

the 5,000-step models reach the strongest returns, and longer training mostly helps the more conservative settings keep returns stable. Low α gives the best peak performance, while moderate and high α keep the policy closer to the logged behavior and preserve ESS. That pattern is consistent with the appendix heatmaps and alpha curves.

The policy support diagnostics tell the same story. In the action distribution analysis, the CQL policy stays much closer to the clinician support than BC, which is reflected in the much smaller clinician versus CQL divergence than clinician versus BC. The heatmaps also show that CQL concentrates mass in the clinically common fluid and vasopressor bins instead of spreading probability over rare edge cases. This is important because it means the improvement is not coming from unsupported actions, but from better ranking of actions that already appear in the data.

Training dynamics reinforce this interpretation. The CQL objective curves flatten after the early discovery phase, while the stronger conservative settings maintain a smoother trajectory over longer runs. In practice, this means that 5,000 steps is enough to capture most of the gain, and additional training mainly matters when the penalty is large enough to protect ESS and prevent drift.

Appendix A contains the non-critical diagnostics: action heatmaps, mortality-deviation analysis, alpha-sensitivity curves, training curves, and the full sweep heatmaps.

5 Conclusion and Future Work

Overall, the study shows that conservative offline RL can match or exceed the clinician baseline when the state is compact, the reward is clinically meaningful, and the policy is evaluated with a stable OPE estimator. The appendix provides the additional diagnostics that support this claim, including the sweep landscape and hyperparameter sensitivity plots. Future work should focus on seed robustness, binning ablations, and stronger off-policy estimators.

These findings have significant implications for clinical deployment. The minimal feature set, dominated by SOFA Total and Lactate, is already routinely collected in ICU settings, making the model deployable without requiring additional data pipelines. The mortality validation (Figure 2) demonstrates that clinician agreement with the AI policy leads to the best outcomes, building trust for human-in-the-loop decision support. Furthermore, the high ESS ensures that the learned policy can be safely audited before deployment, providing a pathway for evidence-based dosing recommendations rather than autonomous action. Taken together, these results highlight the potential for conservative offline RL to enhance clinical decision-making while maintaining safety and interpretability.

6 Contributions and Acknowledgements

We thank Professor Doina Precup, Nishanth Anand, and the course TAs for their guidance and instruction throughout COMP-579.

- **Alex Groiser:** Data engineering, cohort extraction, MDP construction, training and evaluation.
- **Yusuf Hamza:** OPE pipeline, experiment orchestration, and theoretical analysis.
- **Shared:** Report writing, biological interpretation, final validation, and presentation.

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A Extended Figures and Diagnostics

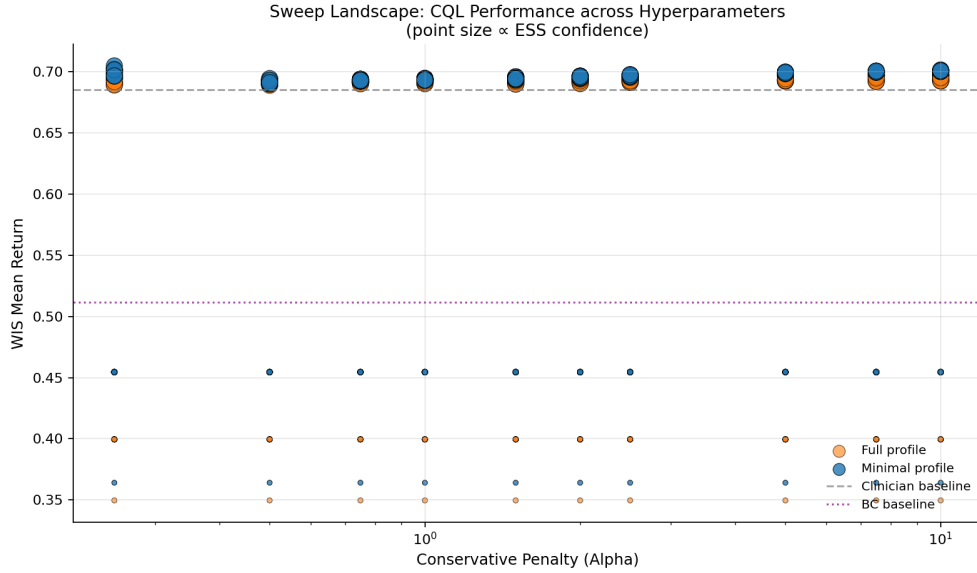


Figure 4: Sweep Landscape: Comprehensive visualization of CQL performance across all hyperparameter configurations. Each point represents one $(\alpha, \text{steps}, \gamma, \text{profile})$ configuration. Point color indicates feature profile (minimal = blue, full = orange), while point size is proportional to the Effective Sample Size (ESS), reflecting our confidence in the OPE estimate. Larger points indicate higher ESS and lower distributional shift, essential for clinical safety.

Extended Sweep Trends and Hyperparameter Analysis

The following figures illustrate the extensive training logs mapping Expected Return (WIS) across 10^5 training steps, partitioned by the feature extraction profile.

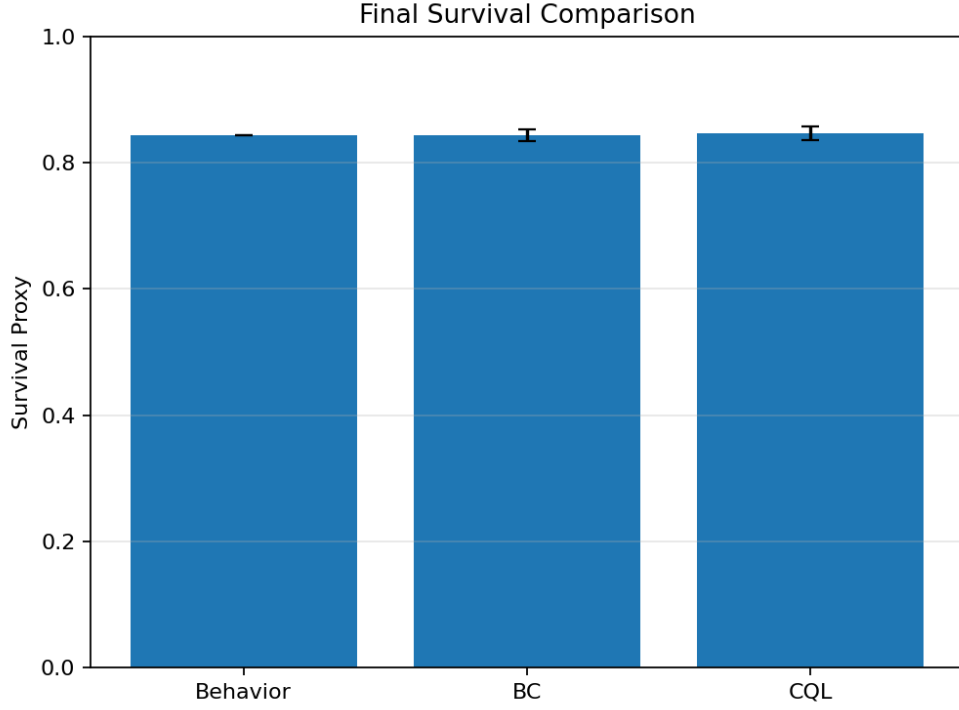


Figure 5: Final Survival Comparison: Bar chart showing the heuristic survival proxy across the evaluated policies.

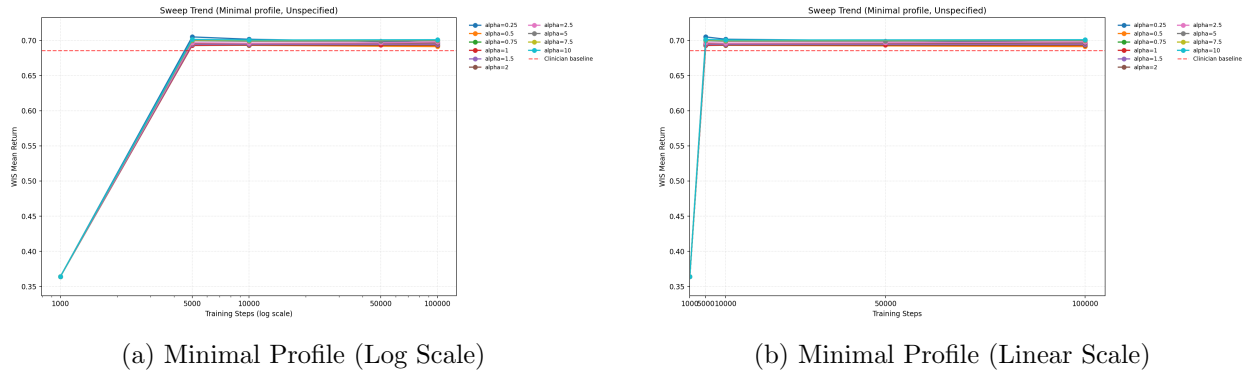
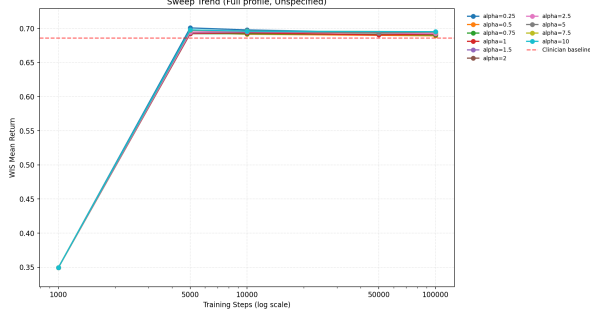
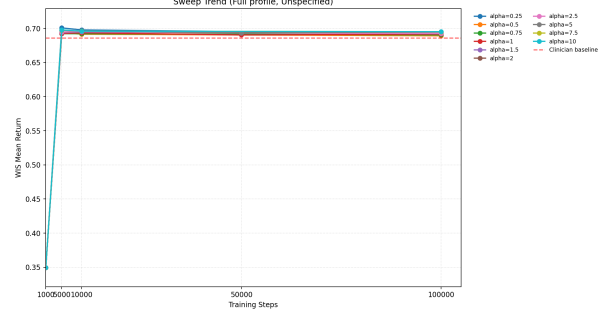


Figure 6: Sepsis Policy Performance comparing convergence stability across various CQL α penalties for the Minimal Profile. Higher conservative penalties successfully lock in stable returns without degrading over extended time horizons.

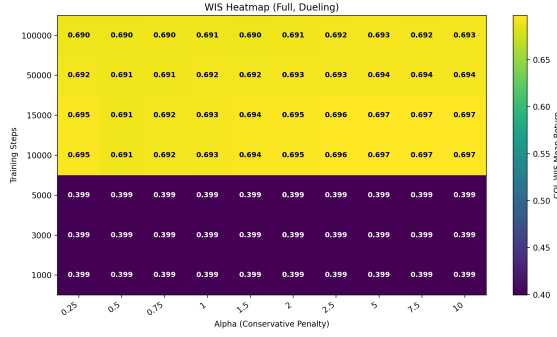


(a) Full Profile (Log Scale)

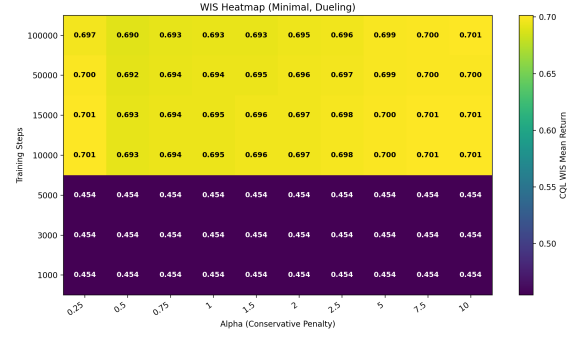


(b) Full Profile (Linear Scale)

Figure 7: Sepsis Policy Performance comparing convergence stability across various CQL α penalties for the Full Profile.



(a) Dueling Architecture Heatmap



(b) Minimal Profile Heatmap

Figure 8: WIS Mean Return heatmaps illustrating the relationship between the conservative penalty α and training steps.

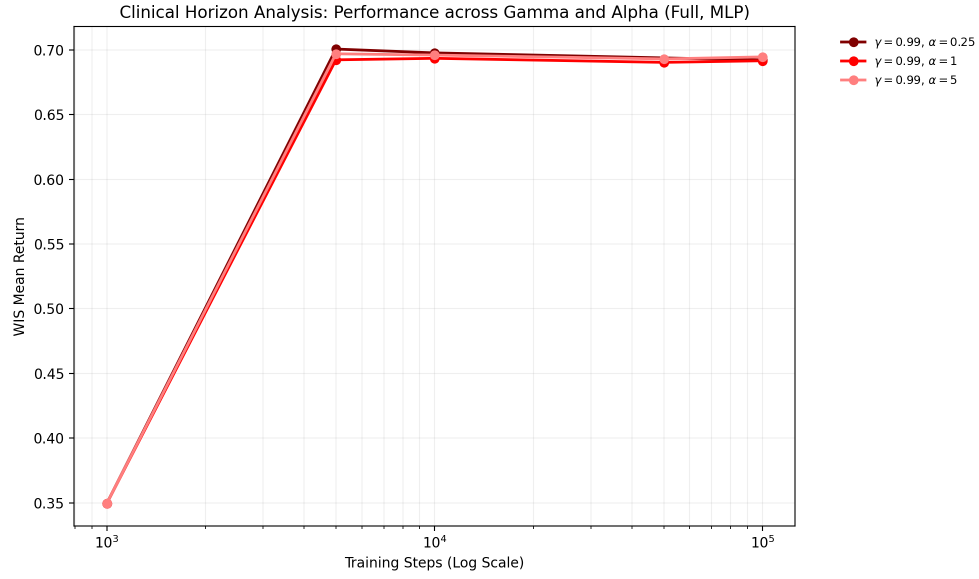


Figure 9: Gamma Sensitivity Analysis: Line chart showing the expected return across different discount factors (γ) for key CQL configurations. However (not shown here) but $\gamma = 0.99$ setting had very good consistent results, and the other gammas under extreme learnign rates had very poor results.

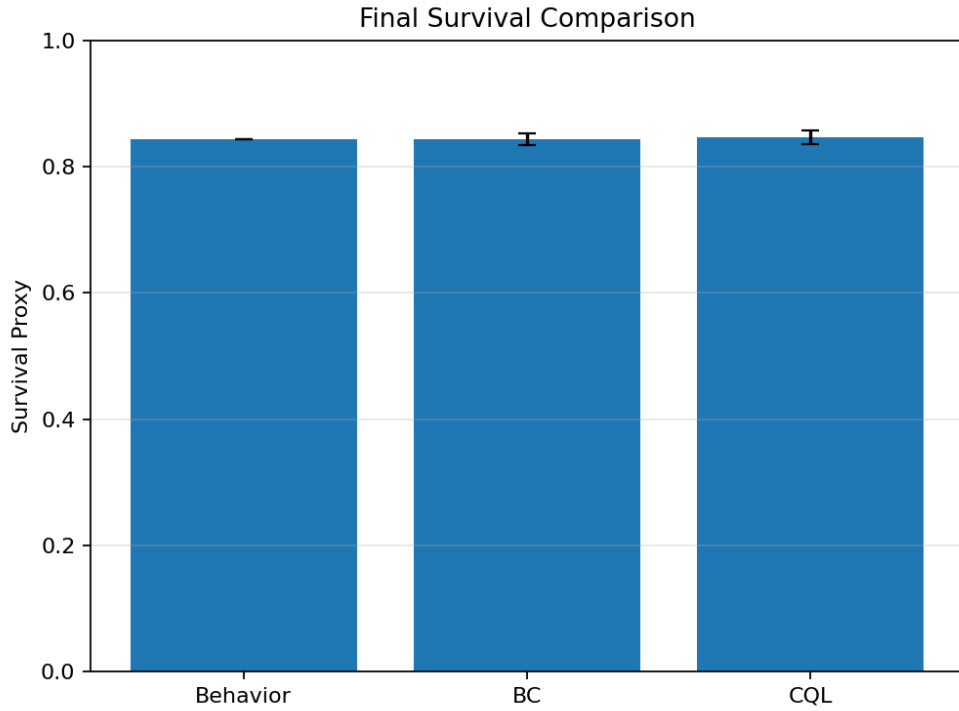


Figure 10: Mean return over alpha (with ci across gammas + steps)